Artificial Neural Networks - Lab Mid

Ablation Study of Regularization Techniques in a Multilayer Perceptron Using the Iris Dataset

Name	Eman Amjad
Roll Number	FA23-BAI-013
Course	Artificial Neural Networks
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Abstract

A 4-64-32-3 Multilayer Perceptron was trained to classify the three iris species. An ablation study then changed one training choice at a time: weight initialization, L2 regularization, dropout and learning rate. The data split, architecture and early-stopping policy were the same for every experiment, and each of the seven experiments (E0-E6) was repeated with three random seeds, giving 21 runs. Trained for 200 epochs without early stopping, the baseline clearly overfits: its validation loss reaches a minimum of 0.145 at epoch 62 and then rises to 0.339, while training loss keeps falling. Early stopping on validation loss removes this effect. With early stopping in place, every configuration except the low learning rate reaches 96.7-98.9% mean test accuracy. L2 and dropout keep the training and validation curves together and make results identical across seeds, but they do not raise test accuracy on this small dataset, and they increase training time by up to ~3×. The learning rate has the largest effect. At 0.0001 the network underfits (test accuracy 0.833, test loss 0.364). At 0.01 it matches the combined model's accuracy with a lower test loss (0.063 vs 0.068) in about one third of the training time.

1. Problem Statement

The goal is to develop a Multilayer Perceptron (MLP) that classifies iris flowers into *Setosa*, *Versicolor* and *Virginica*. The Iris dataset is small and relatively simple, but an MLP with 2,499 parameters trained on only 90 samples for too many epochs can memorize the training data and perform worse on unseen samples.

An ablation study was carried out to measure how individual training choices affect classification performance, generalization and training duration. Starting from a baseline MLP, weight initialization, L2 regularization, dropout and learning rate were varied one at a time, and early stopping ended training once validation performance stopped improving.

2. Experimental Scenario

In preliminary experiments the training accuracy becomes high, but the validation loss starts to rise while the training loss continues to fall, which is the typical sign of overfitting. The study investigates whether regularization, dropout, initialization and learning-rate selection improve generalization. The data split and the architecture are kept fixed so that the comparisons are fair. Section 5.1 reproduces this scenario directly.

3. Dataset and Model Specifications

Property	Specification
Dataset	Iris dataset from scikit-learn
Total samples	150 (50 per class)
Input features	4 (sepal length, sepal width, petal length, petal width; cm)
Output classes	3 (setosa, versicolor, virginica)
Train / validation / test	60% / 20% / 20% (90 / 30 / 30 samples)
Split strategy	Stratified split, fixed random_state = 13 for every experiment (30 / 10 / 10 samples per class in train / val / test)
Preprocessing	StandardScaler fitted on the training set only, then applied to validation and test
Hidden layers	2: 64 and 32 neurons, ReLU, each followed by a Dropout layer (rate p)
Output layer	3 neurons with softmax
Parameters	2,499 trainable (identical in every experiment)
Loss function	Sparse categorical cross-entropy
Optimizer / batch size	Adam / 16
Evaluation metrics	Accuracy, macro F1-score, test loss
Maximum epochs	200
Random seeds	13, 113, 213 (three runs per experiment)

Table 1. Dataset, model and training specifications.

Preventing data leakage.

The scaler's mean and standard deviation are computed from the 90 training samples only and the same fitted scaler transforms the validation and test sets. After scaling, the training features have mean exactly 0 and standard deviation exactly 1. The validation and test means are close to zero but not exactly zero (for example 0.135 and 0.187 for sepal length), which confirms that the scaler never saw those samples.

4. Methodology

4.1 Ablation experiments

Experiment	Initialization	L2	Dropout	Learning rate	Compared with
E0: Baseline	Glorot uniform	0.0	0.0	0.001	-
E1: Initialization	He normal	0.0	0.0	0.001	E0
E2: L2	Glorot uniform	0.01	0.0	0.001	E0
E3: Dropout	Glorot uniform	0.0	0.3	0.001	E0
E4: Combined	Glorot uniform	0.01	0.3	0.001	E0
E5: Low learning rate	Glorot uniform	0.01	0.3	0.0001	E4
E6: High learning rate	Glorot uniform	0.01	0.3	0.01	E4

Table 2. The seven ablation experiments.

E1-E3 each test a single factor against the baseline, E4 combines L2 and dropout, and E5-E6 compare a lower and a higher learning rate against E4. L2 is applied to the kernels of all three Dense layers. The two Dropout layers are always present, and with a rate of 0.0 they pass their input through unchanged, so the architecture is identical in every experiment. Each experiment was repeated with three seeds. A seed fixes the weight initialization, the dropout masks and the mini-batch order, and TensorFlow op-determinism was enabled so that each run is exactly reproducible.

4.2 Early stopping (identical for all experiments)

- Monitor: validation loss (`val_loss`).
- Patience: 15 epochs.
- Restore best weights: True (the reported model is the one from the best epoch).
- Maximum epochs: 200.

4.3 Metrics

For every run the following were recorded: the number of epochs trained, the best epoch, the wall-clock training time, training/validation/test accuracy, test macro F1-score and test loss. The **test loss is the pure cross-entropy** of the restored model on the test set. Keras' `evaluate()` adds the L2 penalty to the loss, which would make the L2 experiments look 3-4× worse for a reason that has nothing to do with prediction quality (both values are given in Appendix A). The **generalization gap** is training accuracy minus test accuracy.

5. Results

5.1 Reproducing the overfitting scenario

To confirm the scenario described in Section 2, the baseline (E0, seed 13) was trained for all 200 epochs **without** early stopping (Figure 1). The validation loss reaches its minimum of **0.145 at epoch 62** and then rises steadily to **0.339** at epoch 200. Over the same epochs the training loss keeps falling, from 0.042 to 0.0061. Training accuracy reaches 100% while validation accuracy ends at 86.7%, below the 96.7% it reached around epoch 30. With patience 15, early stopping would stop this run at epoch 77 and restore the weights from epoch 62.

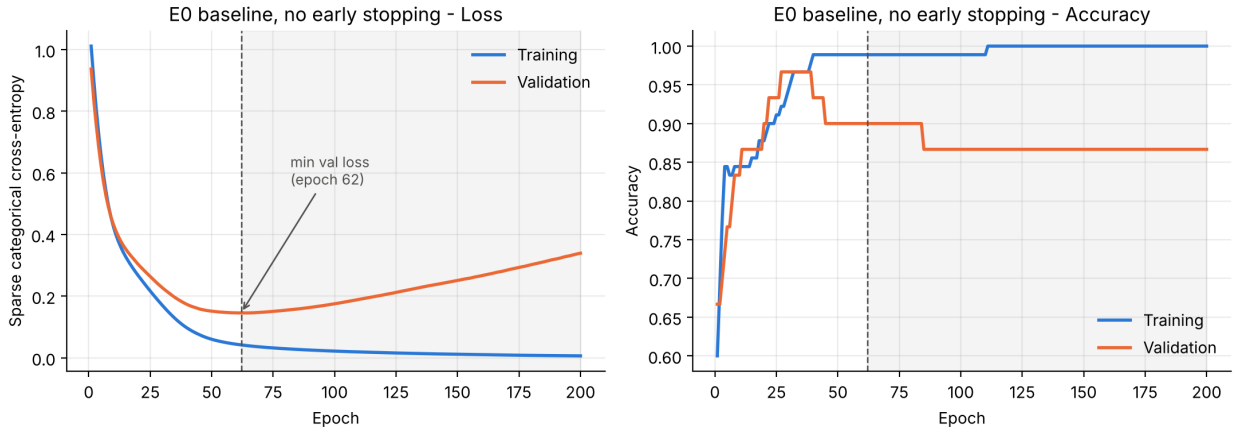


Figure 1. Baseline E0 trained for 200 epochs without early stopping. The dashed line marks the epoch of minimum validation loss; the shaded region is where the model overfits.

5.2 Summary of all experiments

Table 3 gives the mean ± standard deviation over the three seeds. The highest value in each metric column is shown in bold, and the recommended configuration (E6) is highlighted.

Exp.	Test accuracy	Test macro F1	Test loss (CE)	Val CE	Gen. gap	Epochs	Time (s)
E0: Baseline	0.978 ± 0.019	0.978 ± 0.019	**0.053 ± 0.009**	0.162 ± 0.028	0.011 ± 0.019	68.3 ± 7.6	6.9 ± 0.4
E1: Initialization	**0.989 ± 0.019**	**0.989 ± 0.019**	0.066 ± 0.005	0.188 ± 0.019	0.000 ± 0.029	75.3 ± 4.0	7.6 ± 0.4
E2: L2	0.967 ± 0.000	0.967 ± 0.000	0.069 ± 0.002	0.170 ± 0.004	0.022 ± 0.000	200.0 ± 0.0	19.7 ± 0.9
E3: Dropout	0.967 ± 0.000	0.967 ± 0.000	0.054 ± 0.017	**0.153 ± 0.028**	0.019 ± 0.006	99.7 ± 16.5	10.8 ± 1.6
E4: Combined	0.967 ± 0.000	0.967 ± 0.000	0.068 ± 0.004	0.168 ± 0.004	0.022 ± 0.000	189.7 ± 9.3	20.1 ± 1.2
E5: Low learning rate	0.833 ± 0.000	0.822 ± 0.000	0.364 ± 0.028	0.375 ± 0.026	0.015 ± 0.006	200.0 ± 0.0	21.2 ± 0.2
E6: High learning rate	0.967 ± 0.000	0.967 ± 0.000	0.063 ± 0.003	0.157 ± 0.002	0.019 ± 0.006	53.0 ± 18.5	**6.9 ± 1.9**

Table 3. Mean ± standard deviation over seeds 13, 113 and 213. Val CE is the pure validation cross-entropy of the restored model. Training times are wall-clock on a CPU.

5.3 Learning curves

Figure 2 shows the training and validation loss of every experiment for seed 13. The dashed line is the restored (best) epoch. For E2 and E4-E6 the plotted loss is the one Keras optimizes, so it includes the L2 penalty, which is why those curves level off around 0.25-0.35 instead of near zero.

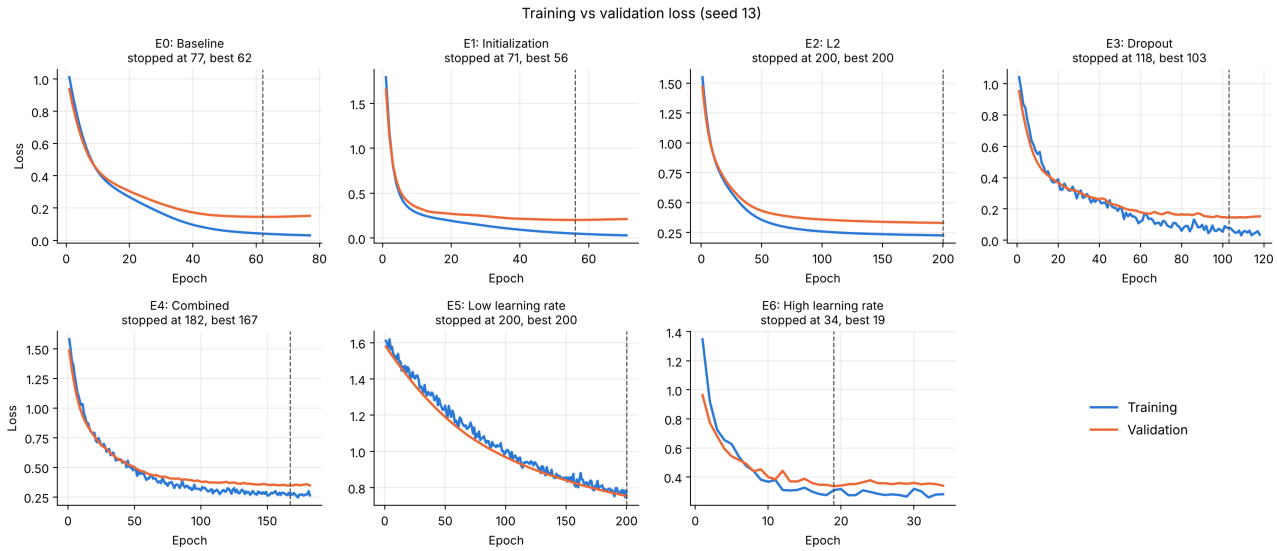


Figure 2. Training vs validation loss for each experiment (seed 13).

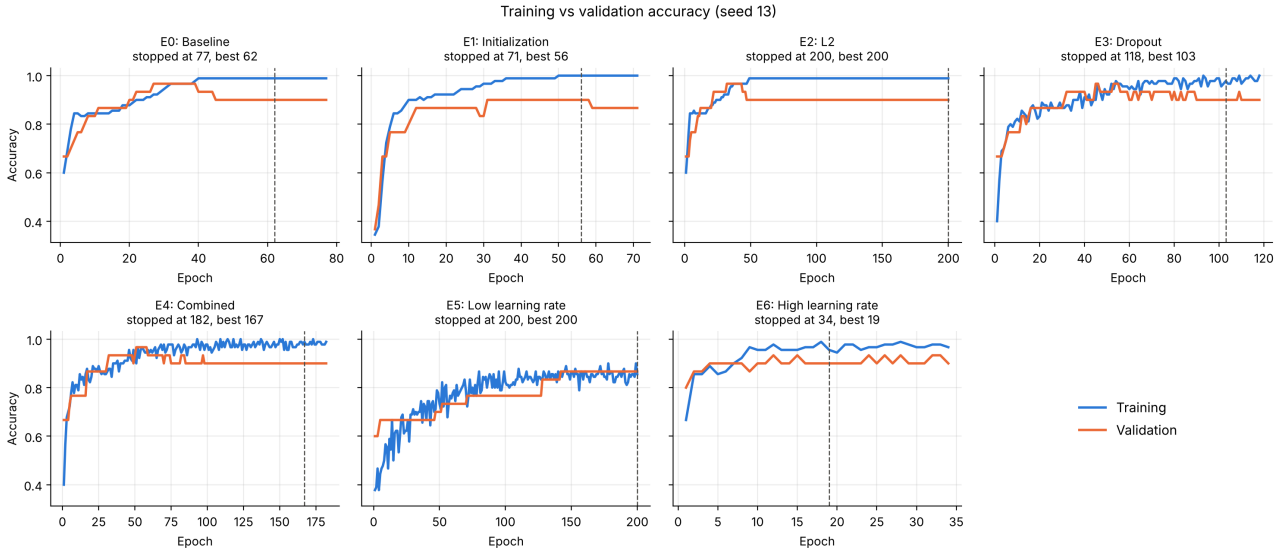


Figure 3. Training vs validation accuracy for each experiment (seed 13).

5.4 Metric comparison

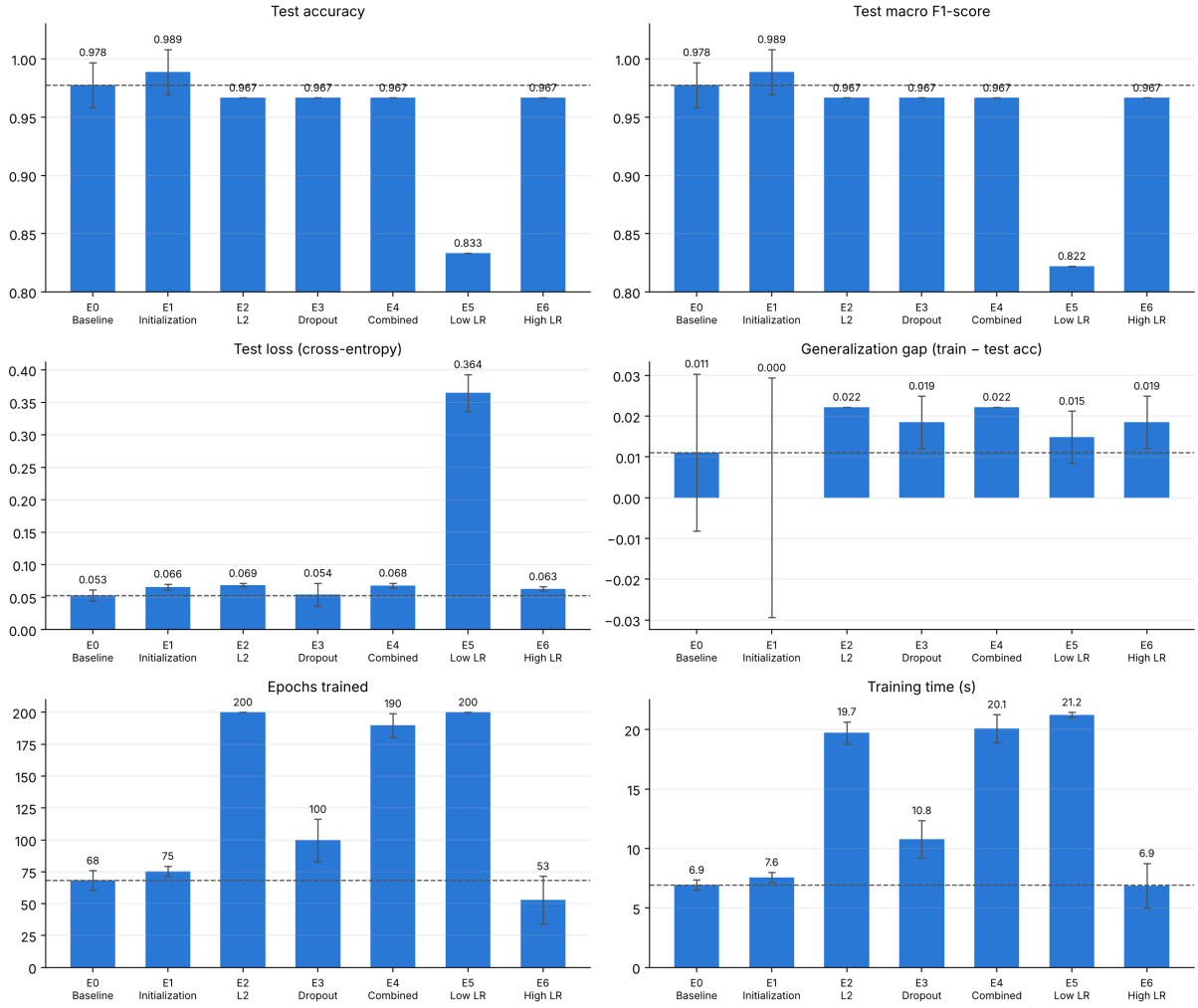


Figure 4. Mean over three seeds (bars) with ± one standard deviation (error bars). The dashed line is the E0 baseline.

5.5 Confusion matrices

Ranked by mean test macro F1 (with lower test loss as the tie-break), the top configuration is E1 (He normal), which classified 89 of the 90 pooled test predictions (3 seeds × 30) correctly, against 88 of 90 for the baseline. In every experiment the only errors are *versicolor* flowers predicted as *virginica*. *Setosa* is linearly separable from the other two classes and is always classified correctly.

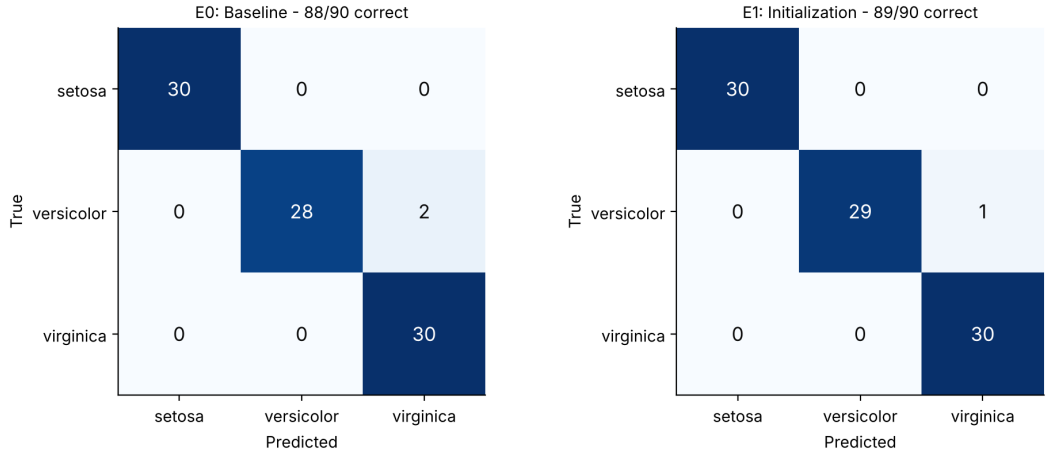


Figure 5. Test-set confusion matrices, three seeds pooled (90 predictions each).

6. Discussion

6.1 Effect of each factor

E1 - He normal initialization. E1 has the highest mean test accuracy and F1 (0.989), but also a *higher* test loss (0.066 vs 0.053) and validation CE (0.188 vs 0.162) than the baseline. The accuracy advantage is one test sample in one seed, which is within seed-to-seed noise. He initialization starts the ReLU layers with a larger weight variance, so the loss falls faster in the first few epochs (Figure 2). It does not generalize better on Iris.

E2 - L2 regularization (λ = 0.01). With L2, test accuracy was 0.967 for **every** seed (std 0), with a test loss of 0.069. The penalty shrinks the weights, which makes the softmax outputs less confident. Training and validation loss run parallel and never diverge, and all three seeds used the full 200 epochs, so training took 2.8× as long as the baseline.

E3 - Dropout (p = 0.3). E3 reached the **lowest validation cross-entropy** of all experiments (0.153) and a test loss equal to the baseline (0.054). Dropout delays overfitting: the best epoch moves from ~53 to ~85. The training curve is noisy because a different random subset of units is dropped at every step.

E4 - L2 + dropout combined. E4 gives the same test accuracy as E2 and E3 (0.967, std 0) and a test loss of 0.068. It is the most stable configuration, with identical predictions for all seeds, but also the slowest of the early-stopped runs (~190 epochs, 20.1 s).

E5 - Low learning rate (0.0001). E5 **underfits**: training accuracy is only 0.848, test accuracy 0.833, macro F1 0.822 and test loss 0.364, roughly 5× worse than E4. Its loss is still decreasing at epoch 200 (Figure 2), so the model ran out of epochs long before converging. The small generalization gap here only means the model performs equally poorly on training and test data.

E6 - High learning rate (0.01). E6 matches E4's accuracy (0.967) with a slightly *lower* test loss (0.063 vs 0.068). It converges in ~53 epochs and 6.9 s, about 3× faster than E4. Its curves are noisier and its stopping epoch varies more between seeds (34-71 epochs).

6.2 Why the L2 runs almost never stop early

E2, E4 and E5 used nearly the whole 200-epoch budget. The `val_loss` that early stopping monitors includes the L2 penalty $\lambda \cdot \Sigma w^2$. As the weights keep shrinking, that penalty keeps falling, so the monitored value keeps improving slightly even after the cross-entropy has levelled off. The numbers confirm this. E4's monitored best validation loss is 0.345, but its pure validation cross-entropy is only 0.168, essentially the same as the baseline's 0.162. About half of what early stopping was watching was the penalty term. This interaction should be kept in mind whenever early stopping is combined with weight decay. Monitoring validation cross-entropy or validation accuracy instead would let these runs stop earlier.

6.3 Overall findings

- **Iris is too easy for accuracy to separate most configurations.** All experiments except E5 classify 29 or 30 of the 30 test samples in every seed, and one sample is worth 3.3 percentage points. Loss values, learning curves and seed-to-seed variance are more informative than accuracy here.
- **Early stopping does most of the work.** It already removes the overfitting seen in Figure 1, which is why the unregularized baseline remains competitive.
- **L2 and dropout change how the model trains more than how well it scores.** They keep the training and validation curves together and remove seed-to-seed variance, and dropout gives the best validation cross-entropy. On a 90-sample, 4-feature problem they do not raise test accuracy, and they cost 1.5-3× more training time.

- **The learning rate has the largest effect of any factor.** At 0.0001 the model underfits badly (-13.3 accuracy points vs E4). At 0.01 it matches E4 in about a third of the time.
- **Recommended configuration: E6** (Glorot uniform, L2 0.01, dropout 0.3, learning rate 0.01). It keeps the regularization that prevents the train/validation curves from diverging, matches E4's accuracy with a lower test loss, and is the fastest to train.

6.4 Limitations

- The validation and test sets contain only 30 samples each, so one flower changes accuracy by 3.3 points. Validation accuracy is exactly 0.900 in almost every run, which suggests three hard versicolor/virginica samples in the validation split.
- Only three seeds and one fixed split were used. Stratified k-fold cross-validation would give tighter error bars.
- The epoch budget and patience are the same for every experiment, as the lab requires. This penalizes E5, which would need many more than 200 epochs to converge at a learning rate of 0.0001.

7. Conclusion

Without early stopping, the baseline MLP overfits the Iris training set: validation loss bottoms out at epoch 62 and then more than doubles while training loss keeps decreasing. Early stopping on validation loss (patience 15, restore best weights) prevents this in every experiment. On top of early stopping, L2 regularization and dropout keep the training and validation curves together and make the results identical across seeds. They do not, however, improve test accuracy on this small and easy dataset (0.967 vs 0.978 for the baseline, a difference of a single test sample). He-normal initialization makes no meaningful difference. The learning rate matters most: 0.0001 underfits (test accuracy 0.833), while 0.01 reaches the same accuracy as 0.001 about 3× faster. Overall, the regularized model with a learning rate of 0.01 (E6) gives the best balance of generalization and training cost.

Appendix A. Results of every run

Exp.	Seed	Epochs	Best ep.	Time (s)	Train acc	Val acc	Test acc	Test F1	Test loss	Keras test loss
E0	13	77	62	7.4	0.9889	0.9000	0.9667	0.9666	0.0442	0.0442
E0	113	63	48	6.6	0.9889	0.9333	1.0000	1.0000	0.0526	0.0526
E0	213	65	50	6.8	0.9889	0.8667	0.9667	0.9666	0.0614	0.0614
E1	13	71	56	7.1	1.0000	0.9000	0.9667	0.9666	0.0687	0.0687
E1	113	76	61	7.8	0.9889	0.9333	1.0000	1.0000	0.0605	0.0605
E1	213	79	64	7.9	0.9778	0.8667	1.0000	1.0000	0.0680	0.0680
E2	13	200	200	18.9	0.9889	0.9000	0.9667	0.9666	0.0695	0.2290
E2	113	200	200	19.5	0.9889	0.9000	0.9667	0.9666	0.0661	0.2278
E2	213	200	200	20.7	0.9889	0.9000	0.9667	0.9666	0.0709	0.2302
E3	13	118	103	12.6	0.9889	0.9000	0.9667	0.9666	0.0417	0.0417
E3	113	95	80	10.0	0.9889	0.9333	0.9667	0.9666	0.0472	0.0472
E3	213	86	71	9.8	0.9778	0.8667	0.9667	0.9666	0.0741	0.0741
E4	13	182	167	18.9	0.9889	0.9000	0.9667	0.9666	0.0688	0.2461
E4	113	200	200	21.2	0.9889	0.9000	0.9667	0.9666	0.0637	0.2376
E4	213	187	172	20.2	0.9889	0.9000	0.9667	0.9666	0.0713	0.2498
E5	13	200	200	21.3	0.8444	0.8667	0.8333	0.8222	0.3752	0.7443
E5	113	200	200	21.4	0.8556	0.8667	0.8333	0.8222	0.3322	0.7206
E5	213	200	200	21.0	0.8444	0.8667	0.8333	0.8222	0.3859	0.7858
E6	13	34	19	4.9	0.9778	0.9000	0.9667	0.9666	0.0648	0.2453
E6	113	54	39	7.2	0.9889	0.9000	0.9667	0.9666	0.0651	0.2351
E6	213	71	56	8.6	0.9889	0.9000	0.9667	0.9666	0.0592	0.2312

Table A1. All 21 runs. "Test loss" is the pure cross-entropy; "Keras test loss" is the value returned by model.evaluate(), which also includes the L2 penalty for E2 and E4-E6.

Appendix B. Reproducibility

- Software: TensorFlow 2.21.0, Keras 3.15.1, scikit-learn 1.9.1.
- Split seed 13; model seeds 13, 113, 213; `tf.config.experimental.enable_op_determinism()` enabled, so every metric except wall-clock time is exactly reproducible.
- The accompanying notebook `LabMid_FA23-BAI-013_Eman_Amjad.ipynb` contains all code, executed outputs and figures. It also writes `results.json`, `results_summary.csv` and `results_per_run.csv`.